METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

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HAZARDOUS-AREA CLASSIFICATION

R-201 synthesis loop · release sources, zones, gas group and temperature class

DOCUMENT NO.	**:**	MEOH-V1-EE-HAC-001
DOCUMENT TITLE	**:**	Hazardous-Area Classification
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Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

1 Purpose, scope and method

This document classifies the hazardous areas around the Methanol Converter Ver 1 synthesis loop (MEOH-PID-001/002) for the selection of electrical and instrument equipment. Method: IEC 60079-10-1 and API 505, quoted as typical practice. The fluids (composition, pressure, temperature) come from the simulator model at design; gas properties are literature values (IEC 60079-20-1 / NFPA 497, stated); the extents are typical values for open-air process plant (assumed), not dispersion calculations.

2 Basis

Item	Basis
Ventilation	Open-air plant, medium degree of ventilation, good availability (assumed). C-201 in an open-sided shelter; analyser shelter mechanically ventilated (≥ 12 air changes/h) with gas detection (assumed).
Grade → zone	Continuous → Zone 0, primary → Zone 1, secondary → Zone 2 (with the ventilation above).
Gas group	IIC wherever hydrogen exceeds 30 vol % of the gas (typical practice, NFPA 497 / API 505): make-up, converter feed, effluent and loop gas are all 73–82 % H2.
Temperature class	From the lowest auto-ignition temperature present: methanol (440 °C) is in the recycle gas → T2 (300 °C); make-up gas without methanol → T1. Selected marking IIC T3 (margin, assumed).
Buoyancy	Loop gas M = 5.83 kg/kmol, relative density 0.201: releases rise; zones are extended upwards. Methanol vapour (1.11) is heavier than air: liquid releases are extended at grade.
Process temperatures	The hottest process surface (catalyst, about 277 °C at design) is below the lowest AIT (440 °C): hot pipework is not an ignition source for these gases.

3 Flammable fluids (from the model)

Fluid	T °C	P bar g	H2 %	CO %	CH4 %	CH3OH %	Inerts %	M	Rel. dens.	LFL %	Min AIT °C	Group	T-class
Make-up syngas (battery limit)	40	86	73	16	3.3	0	7.7	9.74	0.336	4.89	560	IIC	T1
Converter feed / quench (E-202 cold out)	97	79.7	79.99	4.88	9.93	0.29	4.9	6.57	0.227	4.4	440	IIC	T2
Converter effluent (R-201 outlet, E-201, E-202 hot side)	268	77.4	76.9	2.14	10.82	4.78	5.3	7.16	0.247	4.41	440	IIC	T2
Loop gas (V-201 vapour, C-201, purge)	40	75	81.63	2.28	11.48	0.36	4.3	5.83	0.201	4.29	440	IIC	T2
Crude methanol (V-201 liquid, 85.5 wt % CH3OH, dissolved H2/CO2)	40	75	—	—	—	85.5 wt %	water	—	1.11	6	440	IIA	T2

Mixture LFL by Le Chatelier with the inerts (CO2, N2, H2O) as diluent (approximation).

4 Gas properties (literature values, stated)

Component	M	LFL vol %	UFL vol %	AIT °C	Gas group	MESG mm	T-class
Hydrogen (H2)	2.016	4	75	560	IIC	0.29	T1
Carbon monoxide (CO)	28.01	10.9	74	605	IIB	0.84	T1
Methane (CH4)	16.04	4.4	17	600	IIA	1.12	T1
Methanol (CH3OH)	32.04	6	36	440	IIA	0.92	T2

Values as tabulated in IEC 60079-20-1 / NFPA 497; verify against the current edition. T-class limits: T1 450, T2 300, T3 200, T4 135, T5 100, T6 85 °C.

5 Hazardous-area classification table

No.	Equipment	Release source	Grade	Fluid (model)	Zone	Extent (typical, assumed)	Group / T	Ventilation
RS-01	R-201	Flanged nozzles: inlet, outlet, 3 quench, thermowell and PDI tapping flanges, manways, catalyst dump nozzles	Secondary	Converter feed / quench, 79.7 bar g, 97 °C	Zone 2	R 3 m around each flange; buoyant gas: extend 5 m above the top head	IIC T2	Open air, medium ventilation, good availability
RS-02	E-201 / E-202	Channel and shell body flanges, nozzle flanges (both sides 75–80 bar g)	Secondary	Converter effluent, 77.4 bar g, 268 °C	Zone 2	R 3 m	IIC T2	Open air
RS-03	E-203	Process-side flanges; tube leak into cooling water (gas released at the cooling tower)	Secondary	Converter effluent, 77.4 bar g, 268 °C	Zone 2	R 3 m at E-203; cooling tower: see EQ-ee-02	IIC T2	Open air
RS-04	C-201	Shaft seals (dry-gas seal primary vent)	Primary	Loop gas, 75 bar g, 40 °C	Zone 1 / Zone 2	Zone 1 R 1 m around the seal-vent outlet (routed to a safe location); Zone 2 R 4.5 m around the casing	IIC T2	Open-sided shelter, good ventilation assumed
RS-05	C-201	Casing, suction and discharge flanges, lube-oil console vents	Secondary	Loop gas, 75 bar g, 40 °C	Zone 2	R 4.5 m around the machine, 3 m above	IIC T2	Open-sided shelter
RS-06	V-201	Flanges, level bridle (LT-204, LSHH-204), drains	Secondary	Loop gas, 75 bar g, 40 °C	Zone 2	R 3 m	IIC T2	Open air
RS-07	PSV-201	Relief valve body and inlet/outlet flanges (discharge to the closed flare header)	Secondary	Loop gas, 75 bar g, 40 °C	Zone 2	R 3 m around the valve; the flare tip is classified with the flare system	IIC T2	Open air
RS-08	HV-205	Blowdown valve stem packing and flanges (to flare)	Secondary	Loop gas, 75 bar g, 40 °C	Zone 2	R 3 m	IIC T2	Open air
RS-09	AIT-205 / AIT-208	Analyser sample points, sample conditioning and vents (routine release)	Primary	Converter effluent, 77.4 bar g, 268 °C	Zone 1 / Zone 2	Zone 1 R 1 m around the vent and sample point; Zone 2 R 3 m; shelter interior Zone 2 (ventilation ≥ 12 air changes/h with gas detection, assumed)	IIC T2	Shelter vents to a safe location
RS-10	V-201 / LV-204	Crude methanol lab sample point (routine)	Primary	Crude methanol, 75 bar g, 40 °C	Zone 1 / Zone 2	Zone 1 R 1 m; Zone 2 R 3 m, extended at grade for methanol vapour (heavier than air)	IIA T2	Closed sampler recommended
RS-11	LV-204	Let-down valve stem packing and flanges; liquid flashes dissolved H2/CO2 on leaking	Secondary	Crude methanol, 75 bar g, 40 °C	Zone 2	R 3 m, and at grade to the kerb of the paved area	IIA T2	Open air
RS-12	PV-203 / FI-203	Purge valve stem packing, orifice flanges (to the reformer fuel header)	Secondary	Loop gas, 75 bar g, 40 °C	Zone 2	R 3 m	IIC T2	Open air
RS-13	FV-202 / XV-202 / FE-202	Make-up valve stems and flanges at the battery limit	Secondary	Make-up syngas, 86 bar g, 40 °C	Zone 2	R 3 m	IIC T1	Open air
RS-14	E-204	Process-side flanges (HP steam side non-hazardous)	Secondary	Converter feed / quench, 79.7 bar g, 97 °C	Zone 2	R 3 m	IIC T2	Open air

6 Equipment selection

Electrical and instrument equipment in the classified areas: Ex marking IIC T3 (assumed, typical), EPL Gb in Zone 1 (e.g. Ex d, Ex e, Ex ib) and Gc or better in Zone 2. IIC covers every fluid of the unit (IIC hydrogen-rich gases, IIA crude methanol); T3 (200 °C) is below the lowest AIT (440 °C). Field instruments of MEOH-V1-IN-INL-001 are intrinsically safe (Ex ia/ib IIC T4, assumed).

7 Release-rate examples (IEC 60079-10-1 Annex B, choked gas)

Source	Fluid	Hole mm² (Table B.1, typical)	p bar a	W_g g/s	ρ_g kg/m³	W_g/(ρ_g·k·LFL) m³/s
Flange, spiral-wound gasket	Loop gas	0.25	76.0	1.460	0.234	0.1452
Flange, compressed-fibre gasket	Converter feed / quench	2.5	80.7	15.141	0.264	1.3029
C-201 seal (secondary)	Loop gas	2.5	76.0	14.603	0.234	1.4524
Make-up valve stem	Make-up syngas	0.25	87.0	2.161	0.392	0.1128

W_g = C_d·S·p·√(γM/(ZRT)·(2/(γ+1))^((γ+1)/(γ−1))), C_d 0.75, γ 1.40, Z 1.0, k 1.0, ambient 30 °C (assumed). The volumetric release characteristic is what the standard's chart (Annex D) converts into a hazardous distance; the extents of section 5 are typical values and should be confirmed by that method in detail design.

8 Engineering queries

No.	To	Issue	Proposed
EQ-ee-02	MEOH-PID-002 (E-203) / HAZOP	An E-203 tube leak carries hydrogen-rich gas at about 75 bar g into the cooling-water return, which degasses at the cooling tower outside the unit: a release source the unit HAC does not cover.	Add gas detection (or a degassing pot with vent to a safe location) on the E-203 CW return; extend the classification to the cooling-tower return header.
EQ-ee-04	Downstream let-down drum (off-sheet)	LV-204 lets crude methanol down to the let-down drum (about 6 bar g, off-sheet), where dissolved H2 and CO2 flash with methanol vapour: that drum and its vent are not classified here.	Classify the let-down drum, its flash-gas line and the crude-methanol tankage with the downstream unit (HAC rev 1).